

Linear Differential Equations with Constant Coefficients

5.1. Linear Differential Equation

A differential equation of the form

$$\frac{d^n y}{dx^n} + P_1 \frac{d^{n-1} y}{dx^{n-1}} + P_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + P_n y = X$$

where $P_1, P_2, \dots, P_n$, and X are functions of x or constants, is called a *linear differential equation of n^{th} order*.

And if $P_1, P_2, \dots, P_n$ are all constants (not functions of x) and X is some function of x , then the equation is a *linear differential equation with constant coefficients*.

5.2. The Operator D . It is usual to write

$$D \text{ for } \frac{d}{dx}, D^2 \text{ for } \frac{d^2}{dx^2}, \dots, D^n \text{ for } \frac{d^n}{dx^n}.$$

And in terms of the operator D the differential equation (1) can be written as $[D^n + P_1 D^{n-1} + P_2 D^{n-2} + \dots + P_n] y = X$.

Note. It can be proved that D can be treated as an algebraic quantity in several respects.

5.3. A Theorem. If $y=y_1, y=y_2, \dots, y=y_n$ are linearly independent solutions of

$$(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_n) y = 0, \quad \dots (1)$$

then $y = C_1 y_1 + C_2 y_2 + \dots + C_n y_n$ is the general or complete solution of the differential equation, where $C_1, C_2, \dots, C_n$ are n arbitrary constants. ✓

Let us denote the given equation (1) by $f(D) y = 0$, where $f(D) = D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_n$.

Since $y=y_1, y=y_2, \dots, y=y_n$ are solutions of the equation,

$$\therefore f(D) y_1 = 0, f(D) y_2 = 0, \dots, f(D) y_n = 0. \quad \dots (2)$$

Now putting $y = C_1 y_1 + C_2 y_2 + \dots + C_n y_n$ in (1), we have

$$D^n (C_1 y_1 + \dots + C_n y_n) + a_1 D^{n-1} (C_1 y_1 + C_2 y_2 + \dots + C_n y_n) + \dots + a_n (C_1 y_1 + C_2 y_2 + \dots + C_n y_n) = 0$$

$$\text{or } C_1 (D^n y_1 + a_1 D^{n-1} y_1 + \dots + a_n y_1) + C_2 (D^n y_2 + a_1 D^{n-1} y_2 + \dots + a_n y_2) + \dots + C_n (D^n y_n + a_1 D^{n-1} y_n + \dots + a_n y_n) = 0$$

$$\text{or } C_1 f(D) y_1 + C_2 f(D) y_2 + \dots + C_n f(D) y_n = 0$$

$$\text{or } C_1 \cdot 0 + C_2 \cdot 0 + \dots + C_n \cdot 0 = 0 \text{ by (2).}$$

✓ Since (1) is satisfied by $y = C_1 y_1 + C_2 y_2 + \dots + C_n y_n$, it is a solution of (1). Also since it contains n arbitrary constants, it is the general or complete solution of the equation.

5.4. Auxiliary Equation. Consider the differential equation

$$(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_n) y = 0 \quad \dots(1)$$

where $a_1, a_2, \dots, a_n$ are all constants.

Let $y = e^{mx}$ be a solution of this equation. Then putting

$y = e^{mx}$, $Dy = me^{mx}$, $D^2 y = m^2 e^{mx}$, ..., $D^n y = m^n e^{mx}$,
the equation becomes

$$(m^n + a_1 m^{n-1} + a_2 m^{n-2} + \dots + a_n) e^{mx} = 0.$$

Hence e^{mx} will be a solution of (1) if m is a root of the algebraic equation

$$m^n + a_1 m^{n-1} + a_2 m^{n-2} + \dots + a_n = 0 \quad \dots(2)$$

This equation in m is called the *Auxiliary equation*.

Note It is observed that the auxiliary equation $f(m) = 0$ gives the same values of m as the equation $f(D) = 0$ gives of D .

Hence $f(D) = 0$, i.e., $D^n + a_1 D^{n-1} + \dots + a_n = 0$
can in general be regarded as the auxiliary equation.

Therefore in practice we do not replace D by m to form the auxiliary equation. The equation in D may be regarded as auxiliary equation.

5.5. Solution of equation (1) of the above article.

[Gujrat B.Sc. (Prin.) 58; Gujrat B.Sc. (Subsi.) 65]

✓ **Case I.** When all the roots of auxiliary equation are real and different.

If $m_1, m_2, \dots, m_n$ be the n different roots of (2), then $y = e^{m_1 x}, y = e^{m_2 x}, \dots, y = e^{m_n x}$ are all independent solutions of (1). Therefore the general solution of (1) is

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x} + \dots + C_n e^{m_n x}.$$

Ex. 1. ✓ Solve $\frac{d^3 y}{dx^3} - 13 \frac{dy}{dx} - 12y = 0$.

Solution Equation is $(D^3 - 13D - 12)y = 0$.

The auxiliary equation is $(D^3 - 13D - 12) = 0$,
i.e., $(D+1)(D+3)(D-4) = 0$, $D = -1, -3, 4$

Hence the complete solution is

$$y = C_1 e^{-x} + C_2 e^{-3x} + C_3 e^{4x}.$$

Ex. 2. ✓ Solve $(D^3 + 6D^2 + 11D + 6)y = 0$. [Delhi Pass 67]

Solution. A.E. is $(D+1)(D+2)(D+3) = 0$, $D = -1, -2, -3$.

The complete solution is

$$y = C_1 e^{-x} + C_2 e^{-2x} + C_3 e^{-3x}.$$

5.6. Case II. Auxiliary equation having equal roots.

[Gujrat B. Sc. (Princ.) 59; Poona T.D.C. 61 (S)]

We have shown in case I § 5.5, that when $m_1, m_2, \dots, m_n$ are all different, the general solution is

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots + C_n e^{m_n x}.$$

But if $m_1 = m_2$ (two roots equal) then this becomes

$$y = (C_1 + C_2) e^{m_1 x} + C_3 e^{m_3 x} + \dots + C_n e^{m_n x},$$

which clearly contains only $n-1$ arbitrary constants (since $C_1 + C_2$ is equivalent to only one arbitrary constant)

Therefore this is no longer a general solution.

Consider an equation $(D - m_1)^2 y = 0$,
a differential equation of second order having both the roots equal. (1)

Put $(D - m_1) y = v$; then (1) becomes

$$(D - m_1) v = 0 \quad \text{or} \quad \frac{dv}{dx} = m_1 v.$$

Separating the variables, $\frac{dv}{v} = m_1 dx$.

Integrating, $\log v = \log C + m_1 x$ or $v = C e^{m_1 x}$
or $(D - m_1) y = C e^{m_1 x}$ as $v = (D - m_1) y$

$$\text{or} \quad \frac{dy}{dx} - m_1 y = C e^{m_1 x}$$

which is a linear equation of the first order, its I.F. = $e^{-m_1 x}$

$$\therefore y e^{-m_1 x} = \int C e^{m_1 x} e^{-m_1 x} dx = C_2$$

$$\text{or} \quad y = (C_1 + C_2 x) e^{m_1 x}.$$

Therefore the most general solution of

$$(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_n) y = 0,$$

when two roots of A.E. are equal, is

$$y = (C_1 + C_2 x) e^{m_1 x} + C_3 e^{m_3 x} + \dots + C_n e^{m_n x}.$$

Cor. . In case three roots are equal, i.e., $m_1 = m_2 = m_3$, the general solution is

$$y = (C_1 + C_2 x + C_3 x^2) e^{m_1 x} + C_4 e^{m_4 x} + \dots + C_n e^{m_n x}.$$

Ex. 1} Solve $\frac{d^4 y}{dx^4} - \frac{d^3 y}{dx^3} - 9 \frac{d^2 y}{dx^2} - 11 \frac{dy}{dx} - 4y = 0$.

Solution. A.E. is $D^4 - D^3 - 9D^2 - 11D - 4 = 0$,

i.e., $(D+1)^3 (D-4) = 0$, $D = -1, -1, -1, 4$.

Hence the general solution is

$$y = (C_1 + C_2 x + C_3 x^2) e^{-x} + C_4 e^{4x}.$$

Ex. 2. Solve $(D^3 - 2D^2 - 4D + 8) y = 0$.

Solution. Auxiliary equation is

$$D^3 - 2D^2 - 4D + 8 = 0 \quad \text{or} \quad (D+2)(D-2)^2 = 0,$$

$$D = -2, 2, 2.$$

$$\therefore y = (C_1 + C_2 x) e^{-2x} + C_3 e^{2x}.$$

(Delhi Pass 1968)

5.7 Case III. Auxiliary equation having imaginary roots.

Let $\alpha \pm i\beta$ be the imaginary roots of an equation of second order (since imaginary roots occur in pairs).

Then its general solution is

$$\begin{aligned} y &= C_1 e^{(\alpha+i\beta)x} + C_2 e^{(\alpha-i\beta)x} \\ &= e^{\alpha x} [C_1 e^{i\beta x} + C_2 e^{-i\beta x}] \\ &= e^{\alpha x} [C_1 (\cos \beta x + i \sin \beta x) + C_2 (\cos \beta x - i \sin \beta x)] \\ &= e^{\alpha x} [(C_1 + C_2) \cos \beta x + (C_1 - C_2) i \sin \beta x] \\ &= e^{\alpha x} [A \cos \beta x + B \sin \beta x]. \end{aligned}$$

Note. The above result after suitably adjusting constants may also be written as

$$y = e^{\alpha x} \cdot A \cos(\beta x + B) \quad \text{or} \quad y = e^{\alpha x} \cdot A \sin(\beta x + B).$$

Imaginary roots repeated. If auxiliary equation has two equal pairs of imaginary roots, i.e., if $\alpha + i\beta$ and $\alpha - i\beta$ occur twice, then general solution is obtained as

$$y = e^{\alpha x} [C_1 + C_2 x] \cos \beta x + (C_3 + C_4 x) \sin \beta x.$$

Cor. If a pair of roots of the auxiliary equation occur in the form of quadratic surd $\alpha \pm \sqrt{\beta}$, where β is +ve, then the corresponding term in the solution may be written as

$$e^{\alpha x} [C_1 \cosh x\sqrt{\beta} + C_2 \sinh x\sqrt{\beta}]$$

$$\text{or } C_1 e^{\alpha x} \cosh(x\sqrt{\beta} + C_2) \quad \text{or} \quad C_1 e^{\alpha x} \sinh(x\sqrt{\beta} + C_2).$$

Ex. 1. Solve $(D^4 + 5D^2 + 6)y = 0$. (Karnatak M.A. 61)

Solution. Auxiliary equation is $(D^4 + 5D^2 + 6) = 0$,

$$\text{i.e., } (D^2 + 3)(D^2 + 2) = 0 \quad \therefore D = \pm \sqrt{3}i, \pm \sqrt{2}i.$$

Hence the complete solution is

$$y = C_1 \cos \sqrt{3}x + C_2 \sin \sqrt{3}x + C_3 \cos \sqrt{2}x + C_4 \sin \sqrt{2}x.$$

Ex. 2. Solve $(D^4 - D^3 - D + 1)x = 0$ (Gujrat 58)

Solution. Auxiliary equation is $D^4 - D^3 - D + 1 = 0$

$$\text{or } (D^3 - 1)(D - 1) = 0 \quad \text{or } (D - 1)^2 (D^2 + D + 1) = 0$$

$$\text{or } D = 1, 1, -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i.$$

Hence the complete solution is

$$y = (C_1 + C_2 x) e^x + e^{-x/2} \left[C_3 \cos \frac{\sqrt{3}}{2} x + C_4 \sin \frac{\sqrt{3}}{2} x \right].$$

Ex. 3. Solve the differential equation

$$\frac{d^2 y}{dx^2} + a \frac{dy}{dx} + by = 0,$$

a, b being constants.

(Delhi Hons. 66)

Solution. Proceed yourself.

5.8. Synopsis of the forms of solutions

To solve an equation of the form

$$(D^n + a_1 D^{n-2} + a_2 D^{n-2} + \dots + a_n) y = 0 :$$

1. Find the roots of the auxiliary equation, viz.

$$D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_n = 0.$$

2. Put the General Solution as follows :

Roots of Aux. Equation	Complete Solution
Case I All roots $m_1, m_2, m_3, \dots, m_n$ real and different.	$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots + C_n e^{m_n x}$
Case II $m_1 = m_2$ but other roots real and different.	$y = (C_1 + C_2 x) e^{m_1 x} + C_3 e^{m_3 x} + \dots + C_n e^{m_n x}$
Case III (Imag. Roots) 1. $\alpha \pm i\beta$, a pair of imaginary roots.	Corresponding part of the general solution is $e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$ or $C_1 e^{\alpha x} \cos (\beta x + C_2)$ or $C_1 e^{\alpha x} \sin (\beta x + C_2).$
2. $(\alpha \pm i\beta), (\alpha \pm i\beta)$ repeated twice.	Corresponding part of general solution is $y = e^{\alpha x} [(C_1 + C_2 x) \cos \beta x + (C_3 + C_4 x) \sin \beta x].$

Ex. 1 Solve $\frac{d^4 y}{dx^4} - a^4 y = 0.$

Solution. The auxiliary equation is $(D^4 - a^4) = 0$
 or $(D^2 - a^2)(D^2 + a^2) = 0, D = \pm a, \pm ai.$

$\therefore$ solution is $y = C_1 e^{ax} + C_2 e^{-ax} + (C_3 \cos ax + C_4 \sin ax).$

Ex. 2 Solve $\frac{d^4 y}{dx^4} + m^4 y = 0.$

Solution. Auxiliary equation is $D^4 + m^4 = 0$
 or $(D^2 + m^2)^2 - 2m^2 D^2 = 0$
 or $(D^2 - \sqrt{2}mD + m^2)(D^2 + \sqrt{2}mD + m^2) = 0.$

When $D^2 - \sqrt{2}mD + m^2 = 0, D = \frac{m \pm mi}{\sqrt{2}}.$

When $D^2 + \sqrt{2}mD + m^2 = 0, D = \frac{-m \pm mi}{\sqrt{2}}.$

[Agra B. Sc. 55]

i.e., roots of auxiliary equation are $\frac{m}{\sqrt{2}} \pm \frac{m}{\sqrt{2}} i, -\frac{m}{\sqrt{2}} \pm \frac{m}{\sqrt{2}} i$.

Hence the general solution is

$$y = e^{(m/\sqrt{2})x} C_1 \cos\left(\frac{m}{\sqrt{2}}x + C_2\right) + e^{(-m/\sqrt{2})x} C_3 \cos\left(\frac{m}{\sqrt{2}}x + C_4\right).$$

5.9. General solution of $(D^n + a_1 D^{n-1} + \dots + a_n) y = X$ (1)

[Bombay 61: Gujrat 52]

To show that if $y=Y$ is a complete solution of

$$(D^n + a_1 D^{n-1} + \dots + a_n) y = 0 \quad \dots (2)$$

and $y=u$ is a particular solution of (1); then $y=Y+u$ is a general solution of (1), [Nagpur B.Sc. 55 (S)]

Since $y=Y$ is a solution of (2), we have

$$(D^n + a_1 D^{n-1} + \dots + a_n) (Y) = 0. \quad \dots (3)$$

Also since $y=u$ is a solution of (1), we have

$$(D^n + a_1 D^{n-1} + \dots + a_n) u = X. \quad \dots (4)$$

Adding (3) and (4), we have

$$(D^n + a_1 D^{n-1} + \dots + a_n) (Y+u) = X.$$

This shows that $y=Y+u$ is a solution of (1). Now Y being a general solution of (2) contains n arbitrary constants and as such $Y+u$ also contains n arbitrary constants. Therefore $y=Y+u$ is a general solution of (1).

Note 1. In the general solution $y=Y+u$ of the equation (1), Y is called the **Complementary Function (C.F.)** and u is called the **Particular Integral (P. I.)** and thus

The General Solution = C.F. + P.I.

- The solution Y of (2) can be determined by the methods discussed above. The problem is now to find the particular integral u of (1). We give below certain methods of finding u .

Ex. Define the Complementary Function and Particular Integral for the linear differential equation with constant coefficients $f(D)y = X$. [Karnatak 62]

5.10. Meaning of the symbol $\frac{1}{f(D)}$.

Def. $\frac{1}{f(D)} X$ is that function of x , free from arbitrary constants, which when operated by $f(D)$ gives X .

$$\text{Thus } f(D) \cdot \frac{1}{f(D)} X = X.$$

Therefore $f(D)$ and $\frac{1}{f(D)}$ are inverse operators (i.e. they cancel each other's effect on the function on which they operate)

Thus the symbol $\frac{1}{D}$ stands for integration.

✓ 5.11. $\frac{1}{f(D)} X$ is the particular integral of $f(D) y = X$.

Clearly $\frac{1}{f(D)} X$ will be solution of (1) if it satisfies (1).

So putting $\frac{1}{f(D)} X$ for y in (1), we get

$$f(D) \frac{1}{f(D)} X = X \text{ i.e., } X = X, \text{ which is true.}$$

It means that $\frac{1}{f(D)} X$ is a particular solution of (1).

Therefore to find the particular solution of $f(D) y = X$, we should find the value of $\frac{1}{f(D)} X$.

Note. We know that in solving $f(D) y = 0$, $f(D) = 0$ forms the auxiliary equation, which can be resolved into linear factors (real or imaginary). Therefore $\frac{1}{f(D)}$ can be resolved into partial

fractions. The partial fractions will be of the form $\frac{1}{D - \alpha}$ where α is real or imaginary.

5.12. To show that $\frac{1}{D - \alpha} X = e^{\alpha x} \cdot \frac{1}{D} (e^{-\alpha x} X)$.

Suppose $y = \frac{1}{D - \alpha} X$; then $(D - \alpha) y = X$.

or $\frac{dy}{dx} - \alpha y = X$; this is linear in y , as $D \equiv \frac{d}{dx}$.

$\therefore$ Integrating factor $= e^{\int -\alpha dx} = e^{-\alpha x}$

and the solution is $ye^{-\alpha x} = \int e^{-\alpha x} X dx$.

(constant is not added as it is the particular solution)

or $y = e^{\alpha x} \int e^{-\alpha x} X dx$

$$= e^{\alpha x} \frac{1}{D} (X e^{-\alpha x}) \text{ as } \frac{1}{D} \equiv \text{integration.}$$

5.13. Working rule for finding the Particular integral of $f(D) y = X$.

Let $f(D) = (D - \alpha_1)(D - \alpha_2) \dots (D - \alpha_n)$.

Then resolving into partial fraction, we get

$$\frac{1}{f(D)} = \frac{A_1}{D-\alpha_1} + \frac{A_2}{D-\alpha_2} + \dots + \frac{A_n}{D-\alpha_n} \text{ say.}$$

Now particular integral

$$\begin{aligned} &= \frac{1}{f(D)} X = \left\{ \frac{A_1}{D-\alpha_1} + \frac{A_2}{D-\alpha_2} + \dots + \frac{A_n}{D-\alpha_n} \right\} X \\ &= A_1 \frac{1}{D-\alpha_1} X + A_2 \frac{1}{D-\alpha_2} X + \dots + A_n \frac{1}{D-\alpha_n} X \\ &= A_1 e^{\alpha_1 x} \int e^{-\alpha_1 x} X dx + A_2 e^{\alpha_2 x} \int e^{-\alpha_2 x} X dx + \dots \\ &\quad + A_n e^{\alpha_n x} \int e^{-\alpha_n x} X dx. \end{aligned}$$

which can in general be evaluated and thus the particular integral can be found.

Particular Integral in some special cases.

5.14. Particular Integral when $X = e^{ax}$

[Nagpur 61; Poona 61; Karnatak 61; Gujarat 59; Bombay 61]

By successive differentiation, we find that

$$e^{ax} = e^{ax}, \quad \dots (1)$$

$$D e^{ax} = a e^{ax}, \quad \dots (2)$$

$$D^2 e^{ax} = a^2 e^{ax}, \quad \dots (3)$$

$$\dots \dots \dots$$

$$D^n e^{ax} = a^n e^{ax}, \quad \dots (n)$$

If $f(D) = (D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_{n-1} D + a_n)$, then multiplying (1), (2), (3).....(n) by $a_n, a_{n-1}, \dots, 1$ respectively and adding, we obtain

$$f(D) e^{ax} = f(a) e^{ax}$$

Now operating on both the sides by $\frac{1}{f(D)}$,

$$\frac{1}{f(D)} f(D) e^{ax} = \frac{1}{f(D)} f(a) e^{ax}$$

$$\text{or } e^{ax} = f(a) \frac{1}{f(D)} e^{ax} \text{ or } \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax},$$

dividing by $f(a) \neq 0$

$$\boxed{\text{Therefore } \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax}, \text{ provided that } f(a) \neq 0.}$$

$$\text{Ex. 1. Solve } \frac{d^2 y}{dx^2} - 2k \frac{dy}{dx} + k^2 y = 0.$$

[Nagpur 1957]

Solution. Auxiliary equation is $D^2 - 2kD + k^2 = 0$,
i.e., $(D-k)^2 = 0$ or $D = k, k$.

$$\therefore \text{C.F.} = (C_1 + C_2 x) e^{kx}.$$

$$P. I. = \frac{1}{D^2 - 2kD + k^2} e^x = \frac{1}{1 - 2k + k^2} e^x = \frac{1}{(1-k)^2} e^x.$$

Hence the general solution is

$$y = (C_1 + C_2 x) e^{kx} + \frac{1}{(1-k)^2} e^x, k \neq 1.$$

5.15. To show that $\frac{1}{f(D^2)} \sin ax = \frac{1}{f(-a^2)} \sin ax$,
except when $f(-a^2) = 0$. [Poona 1964 ; Delhi 55]

By successive differentiation, we get

$$\sin ax = \sin ax, \quad \dots(1)$$

$$D \sin ax = a \cos ax, \quad \dots(2)$$

$$D^2 \sin ax = -a^2 \sin ax,$$

$$D^3 \sin ax = -a^3 \cos ax,$$

$$D^4 \sin ax = a^4 \sin ax$$

$$\text{or } (D^2)^2 \sin ax = (-a^2)^2 \sin ax, \quad \dots(3)$$

$$\dots \dots \dots \dots \dots$$

$$\text{Similarly } (D^2)^n \sin ax = (-a^2)^n \sin ax.$$

$$\text{Thus } f(D^2) \sin ax = f(-a^2) \sin ax.$$

Operating by $\frac{1}{f(D^2)}$ on both the sides, we get

$$\frac{1}{f(D^2)} f(D^2) \sin ax = \frac{1}{f(D^2)} f(-a^2) \sin ax$$

$$\text{i.e., } \sin ax = f(-a^2) \cdot \frac{1}{f(D^2)} \sin ax.$$

Dividing by $f(-a^2)$, we get

$$\frac{1}{f(D^2)} \sin ax = \frac{1}{f(-a^2)} \sin ax, \text{ if } f(-a^2) \neq 0.$$

$$\text{Similarly } \frac{1}{f(D^2)} \cos ax = \frac{1}{f(-a^2)} \cos ax.$$

Important. It follows from the result above that we put $-a^2$ in place of D^2 . We cannot put anything in place of D .

Thus for D^2 put $-a^2$.

for $D^3 = D^2 \cdot D$ put $-a^2 D$.

for $D^4 = D^2 \cdot D^2$ put $-a^2 (-a^2)$, i.e., a^4 etc.

Thus ultimately $f(D)$ becomes linear in D say of the form $(D + \alpha)$. Then we proceed as follows :

$$\frac{1}{D + \alpha} \sin ax = \frac{(D - \alpha)}{(D + \alpha)(D - \alpha)} \sin ax$$

$$= \frac{(D - \alpha)}{D^2 - \alpha^2} \sin ax = \frac{D - \alpha}{-a^2 - \alpha^2} \sin ax$$

putting $-a^2$ for D^2 in the denominator

$$= \frac{1}{-a^2 - \alpha^2} \left(\frac{d}{dx} \sin ax - \alpha \sin ax \right) \text{ as } D = \frac{d}{dx}$$

$$= \frac{1}{-a^2 - \alpha^2} (a \cos ax - \alpha \sin ax),$$

And thus the particular integral in case of $\sin ax$ and $\cos ax$ can be completely evaluated.

Ex. 1. Solve $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = \sin 2x$.

[Calcutta Hons 1962 ; Karnatak 60 ; Saugar 59 ;
Raj. 59 ; Gujrat 61]

Solution. A.E. is $D^2 + D + 1 = 0$, $D = -\frac{1}{2} \pm \frac{1}{2}\sqrt{3}i$.

$\therefore$ C.F. $= e^{-(1/2)x} C_1 \cos \{\frac{1}{2}\sqrt{3}x + C_2\}$.

$$\text{P.I.} = \frac{1}{D^2 + D + 1} \sin 2x = \frac{1}{-4 + D + 1} \sin 2x$$

$$= \frac{1}{D - 3} \sin 2x = \frac{D + 3}{D^2 - 9} \sin 2x$$

$$= -\frac{1}{13} (2 \cos 2x + 3 \sin 2x).$$

Hence the complete solution is

$$y = e^{-x/2} C_1 \cos \left\{ \frac{1}{2}\sqrt{3}x + C_2 \right\} - \frac{1}{13} (2 \cos 2x + 3 \sin 2x).$$

Ex. 2. Solve $(D^2 + 1)^2 y = \cos 3x$.

[Bombay 1958]

Solution. A.E. is $(D^2 + 1)^2 = 0$, $D = \pm i, \pm i$.

$\therefore$ C.F. $= (C_1 + C_2 x) \cos x + (C_3 + C_4 x) \sin x$.

$$\text{P.I.} = \frac{\cos 3x}{(D^2 + 1)^2} = \frac{\cos 3x}{(-9 + 1)^2} = \frac{1}{8} \cos 3x.$$

Hence the complete solution is

$$y = (C_1 + C_2 x) \cos x + (C_3 + C_4 x) \sin x + \frac{1}{8} \cos 3x.$$

Ex. 3. Solve $(D^3 + D^2 + D + 1)y = \sin 2x$.

[Poona 1963]

Solution. A.E. is $(D^2 + 1)(D + 1) = 0$, $D = -1, \pm i$.

C.F. $= C_1 e^{-x} + C_2 \cos x + C_3 \sin x$ or $C_1 e^{-x} + C_2 \cos x + C_3 \sin x$.

$$\text{P.I.} = \frac{1}{(D^2 + 1)(D + 1)} \sin 2x = \frac{1}{(-4 + 1)(D + 1)} \sin 2x$$

$$= -\frac{1}{3} \frac{D - 1}{D^2 - 1} \sin 2x = -\frac{1}{3} \frac{D - 1}{-4 - 1} \sin 2x$$

$$= \frac{1}{15} [2 \cos 2x - \sin 2x].$$

Complete solution is $y = \text{C.F.} + \text{P.I.}$

Ex. 4. Prove that the solution of the differential equation $\frac{d^2y}{dx^2} + 4y = \sin ax$ when $a \neq 2$, under the conditions $y = 0$ and $\frac{dy}{dx} = 0$ when $x = 0$ is $y = \frac{2 \sin ax - a \sin 2x}{2(4 - a^2)}$.

[Nagpur 1961]

Solution. A.E. is $D^2 + 4 = 0$, $D = \pm 2i$.

$$C.F. = C_1 \sin (2x + C_2).$$

$$P.I. = \frac{1}{D^2 + 4} \sin ax = \frac{\sin ax}{4 - a^2}.$$

$\therefore$ The general solution is

$$y = C_1 \sin (2x + C_2) + \frac{\sin ax}{4 - a^2}. \quad \dots(1)$$

$$\text{so that } \frac{dy}{dx} = 2C_1 \cos (2x + C_2) + \frac{a \cos ax}{4 - a^2}. \quad \dots(2)$$

But $y=0$ when $x=0$,

$$\therefore (1) \text{ gives } 0 = C_1 \sin C_2. \quad \dots(3)$$

$$\text{Again } \frac{dy}{dx} = 0 \text{ when } x=0.$$

$$\therefore (2) \text{ gives } 0 = 2C_1 \cos C_2 + \frac{a}{4 - a^2}. \quad \dots(4)$$

From (3), $C_1=0$ or $C_2=0$ but if $C_1=0$, (4) does not hold.

$$\text{Hence } C_2=0 \text{ and then from (4), } C_1 = -\frac{a}{2(4 - a^2)}.$$

Putting these values of C_1 and C_2 in (1), the required solution is

$$y = -\frac{a \sin 2x}{2(4 - a^2)} + \frac{\sin ax}{4 - a^2} = \frac{2 \sin ax - a \sin 2x}{2(4 - a^2)}.$$

This proves the result.

5.16. Exceptional case of $\frac{1}{f(D)} e^{ax}$ when $f(a)=0$.

[Poona 61 ; Bombay 61]

We have from 5.14, $\frac{1}{f(D)} e^{ax} = \frac{e^{ax}}{f(a)}$ if $f(a) \neq 0$.

But if $f(a)=0$, this becomes infinite and our method fails.

Now $f(a)=0$ means that $(D-a)$ is a factor of $f(D)$.

Therefore let $f(D) = (D-a) \phi(D)$,

such that $\phi(a) \neq 0$.

$$\therefore \frac{1}{f(D)} e^{ax} = \frac{1}{(D-a) \phi(D)} e^{ax} \quad \dots(1)$$

$$= \frac{1}{D-a} \cdot \frac{1}{\phi(D)} e^{ax} \text{ as } \phi(a) \neq 0$$

$$= \frac{1}{\phi(a)} \frac{1}{D-a} e^{ax} = \frac{1}{\phi(a)} \cdot e^{ax} \int e^{-ax} e^{ax} dx \quad [\text{§ 5.12}]$$

$$= \frac{1}{\phi(a)} e^{ax} \int dx = \frac{x e^{ax}}{\phi(a)}.$$

Now differentiating both the sides of (1) w.r.t. D , ... (2)

$$f'(D) = (D-a) \phi'(D) + \phi(D).$$

$$\text{Putting } D=a, \quad f'(a) = 0 + \phi(a).$$

It means $\phi(a) = f'(a)$.

Hence (2) becomes

$$\frac{1}{f(D)} e^{ax} = \frac{x e^{ax}}{f'(a)} \quad \text{or} \quad x \cdot \frac{1}{f'(D)} e^{ax}.$$

Again if $f'(a) = 0$ and $f''(a) \neq 0$ then $D - a$ is a factor repeated twice; and applying the above result once again, we get

$$\frac{1}{f(D)} e^{ax} = x^2 \frac{1}{f''(D)} e^{ax} \text{ and so on.}$$

*5.17. Exceptional case of $\frac{1}{f(D^2)} \sin ax$ when $f(-a^2) = 0$.

[Delhi Hons. 65, 64]

From § 5.15 P. 68, $\frac{1}{f(D^2)} \sin ax = \frac{1}{f(-a^2)} \sin ax$, $f(-a^2) \neq 0$.

But if $f(-a^2) = 0$, it becomes infinite and our method fails.

Now $f(-a^2) = 0$ means that $D^2 + a^2$ is a factor of $f(D^2)$.

Let $f(D^2) = (D^2 + a^2) \phi(D^2)$, such that $\phi(-a^2) \neq 0$.

$$\text{Now } \frac{1}{f(D^2)} (\cos ax + i \sin ax) = \frac{1}{f(D^2)} e^{aix}$$

$$= x \frac{1}{f'(D^2)} e^{aix}$$

where dashes denote differentiation w.r.t. D

$$= x \frac{1}{f'(D^2)} (\cos ax + i \sin ax).$$

Equating real and imaginary parts, we have

$$\frac{1}{f(D^2)} \cos ax = x \frac{1}{f'(D^2)} \cos ax$$

and $\frac{1}{f(D^2)} \sin ax = x \frac{1}{f'(D^2)} \sin ax.$

In case $f'(-a^2) = 0$ and $f''(-a^2) \neq 0$, $D^2 + a^2$ is a twice repeated factor of $f(D^2)$. Applying the above result once again, we get

$$\frac{1}{f(D^2)} \sin ax = x^2 \frac{1}{f''(D^2)} \sin ax$$

and $\frac{1}{f(D^2)} \cos ax = x^2 \frac{1}{f''(D^2)} \cos ax$

Ex. 1. Solve $\frac{d^2y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^x$

Solution. Auxiliary equation is

$$D^2 - 3D + 2 = 0 \quad \text{i.e., } (D-2)(D-1) = 0.$$

$$\therefore \text{C.F. } C_1 e^x + C_2 e^{2x}.$$

P.I. = $\frac{e^x}{D^2 - 3D + 2}$ (case of failure)

[Karnatak 60]

$$= x \frac{e^x}{2D-3} \text{ multiplying by } x \text{ and}$$

differentiating the deno. w.r.t. D .

$$= x \frac{e^x}{2 \cdot 1 - 3} = -x e^x.$$

Hence the complete solution is $y = C_1 e^x + C_2 e^{2x} - x e^x$.

Ex. 2. Solve $(D^2 + 4D + 3)y = e^{-3x}$.

[Gujrat 61]

Solution. Auxiliary equation is

$$D^2 + 4D + 3 = 0, (D+3)(D+1) = 0.$$

$$\therefore \text{C. F.} = C_1 e^{-x} + C_2 e^{-3x}.$$

$$\text{P. I.} = \frac{e^{-3x}}{D^2 + 4D + 3}, \text{ case of failure}$$

$$= x \frac{e^{-3x}}{2D+4} \text{ multiplying by } x \text{ and differentiating the denominator w.r.t. } D$$

$$= x \frac{e^{-3x}}{2(-3)+4} = -\frac{1}{2} x e^{-3x}.$$

Hence the general solution is

$$y = C_1 e^{-x} + C_2 e^{-3x} - \frac{1}{2} x e^{-3x}.$$

Ex. 3. Solve $\frac{d^3 y}{dx^3} + 3 \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + y = e^{-x}$.

[Gujrat 61]

Solution. Auxiliary equation is

$$D^3 + 3D^2 + 3D + 1 = 0, (D+1)^3 = 0, D = -1, -1, -1.$$

$$\therefore \text{C. F.} = (C_1 + C_2 x + C_3 x^2) e^{-x}.$$

$$\text{P. I.} = \frac{e^{-x}}{(D+1)^3} \text{ (case of failure)}$$

$$= x \frac{e^{-x}}{3(D+1)^2} \text{ multiplying by } x \text{ and differentiating the denominator w.r.t. } D \text{ (this is again a case of failure)}$$

$$= x^2 \frac{e^{-x}}{6(D+1)} \text{ multiplying again by } x \text{ and differentiating the denominator w.r.t. } D \text{ (again case of failure)}$$

$$= x^3 \frac{e^{-x}}{6} \text{ multiplying by } x \text{ again and differentiating the denominator w.r.t. } D.$$

Hence the complete solution is

$$y = (C_1 + C_2 x + C_3 x^2) e^{-x} + \frac{1}{6} x^3 e^{-x}.$$

Ex. 4. Solve $2 \frac{d^3 y}{dx^3} - 3 \frac{d^2 y}{dx^2} + y = e^x + 1$.

Solution. Auxiliary equation is $2D^3 - 3D^2 + 1 = 0$
or $(D-1)(D-1)(2D+1) = 0, D = 1, 1, -\frac{1}{2}$

$$\text{C. F.} = (C_1 + C_2 x) e^x + C_3 e^{-x/2}$$

[Poona 61]

$$\begin{aligned} \text{P. I.} &= \frac{e^x}{2D^3 - 3D^2 + 1} + \frac{1}{2D^3 - 3D^2 + 1} \text{ first term case of failure} \\ &= x \frac{e^x}{6D^2 - 6D} + \frac{e^{2x}}{0 - 3 \cdot 0 + 1} \text{ differentiating the denominator} \\ &\quad \text{of the first and multiplying it} \\ &\quad \text{by } x \text{ (again case of failure)} \\ &= x^2 \frac{e^x}{12D - 6} + 1 \text{ again differentiating the denominator} \\ &\quad \text{and multiplying by } x. \\ &= \frac{1}{6} x^2 e^x + 1. \end{aligned}$$

Hence the complete solution is

$$y = (C_1 + C_2 x) e^x + C_3 e^{-x/2} + \frac{1}{6} x^2 e^x + 1.$$

Ex. 5. $(D^3 - 2D^2 - 5D + 6)y = e^{3x}$.

[Poona 63]

Solution. A. E. is $(D - 3)(D^2 + D - 2) = 0$.

i.e., $(D - 3)(D + 2)(D - 1) = 0$.

$\therefore$ C. F. $= C_1 e^x + C_2 e^{-2x} + C_3 e^{3x}$.

$$\begin{aligned} \text{P. I.} &= \frac{1}{(D - 3)(D + 2)(D - 1)} e^{3x} = \frac{1}{(D - 3)(3 + 2)(3 - 1)} e^{3x} \\ &= \frac{1}{10(D - 3)} e^{3x} \\ &= x \cdot \frac{1}{10} e^{3x} \text{ as it is a case of failure.} \end{aligned}$$

$\therefore$ The complete solution is $y = \text{C. F.} + \text{P. I.}$

Ex. 6. (a) Solve $\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 4y = e^{2x} + e^{-2x}$.

[Karnatak 60]

Solution. A. E. is $(D^2 + 4D + 4) = 0$, i.e., $(D + 2)^2 = 0$.

$\therefore$ C. F. $= (C_1 + C_2 x) e^{-2x}$.

$$\begin{aligned} \text{P. I.} &= \frac{e^{2x}}{(D + 2)^2} + \frac{e^{-2x}}{(D + 2)^2} \text{ second is a case of failure} \\ &= \frac{e^{2x}}{(2 + 2)^2} + \frac{x^2 e^{-2x}}{2} \text{ differentiating denominator of the} \\ &\quad \text{second twice w.r.t. } D \text{ and multiplying by } x^2 \\ &= \frac{e^{2x}}{16} + \frac{1}{2} x^2 e^{-2x}. \end{aligned}$$

Hence the complete solution is $y = \text{C. F.} + \text{P. I.}$

Ex. 6. (b) Solve $\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 4y = 2 \sinh 2x$.

[Delhi Hons. 62]

Hint. $2 \sinh 2x = e^{2x} - e^{-2x}$. Now proceed as in Ex. 6 (a).

Ex. 6. (c) Solve the following :

$$\frac{d^2 x}{dt^2} = x + e^t + e^{-t}.$$

[Delhi Pass 68]

Solution. Auxiliary equation is

$$D^2 - 1 = 0, \text{ i.e., } D = \pm 1$$

($D \equiv d/dt$).

$$\therefore \text{C.F.} = C_1 e^{-t} + C_2 e^t$$

$$\text{P.I.} = \frac{e^t + e^{-t}}{(D^2 - 1)} = \frac{e^t}{D^2 - 1} + \frac{e^{-t}}{D^2 - 1} \text{ exceptional cases}$$

$$= t \frac{e^t}{2D} + t \frac{e^{-t}}{2D}$$

$$= \frac{1}{2} t e^t - \frac{1}{2} t e^{-t}.$$

$\therefore x = C_1 e^{-t} + C_2 e^t + \frac{1}{2} t (e^t - e^{-t})$ is the general solution.

Ex. 7. Solve $(D^2 + a^2)y = \sin ax$. [Poona 62; Saugar 63]

Solution. A. E. is $D^2 + a^2 = 0$, $D = \pm ai$.

$$\therefore \text{C.F.} = C_1 \cos(ax + C_2).$$

$$\text{P.I.} = \frac{\sin ax}{D^2 + a^2} \text{ case of failure}$$

$$= x \frac{\sin ax}{2D} \text{ multiplying by } x \text{ and differentiating the denominator w.r.t. } D$$

$$= -\frac{x}{2a} \cos ax.$$

Hence $y = C_1 \cos(ax + C_2) - \frac{x}{2a} \cos ax$ is the complete solutions.

Ex. 8. Solve $\frac{d^4 y}{dx^4} + y = \sin \frac{3}{2}x \sin \frac{1}{2}x$.

[Raj. 61]

Solution. Auxiliary equation is

$$D^4 + 1 = 0 \text{ or } (D^2 + 1)(D^2 - D^2 + 1) = 0$$

$$\text{or } (D^2 + 1)[(D^2 + 1)^2 - 3D^2] = 0$$

$$\text{or } (D^2 + 1)(D^2 - \sqrt{3}D + 1)(D^2 + \sqrt{3}D + 1) = 0.$$

$$\text{When } D^2 + 1 = 0, \quad D = \pm i.$$

$$\text{When } D^2 - \sqrt{3}D + 1 = 0, \quad D = \frac{\sqrt{3} \pm i}{2}.$$

$$\text{When } D^2 + \sqrt{3}D + 1 = 0, \quad D = \frac{-\sqrt{3} \pm i}{2}.$$

$$\text{Hence C.F.} = C_1 \cos(x + C_2) + C_3 e^{i\sqrt{3}x} \cos(\frac{1}{2}x + C_4) + C_5 e^{-i\sqrt{3}x} \cos(\frac{1}{2}x + C_6).$$

$$\text{Now } \sin \frac{3}{2}x \sin \frac{1}{2}x = \frac{1}{2} (\cos x - \cos 2x).$$

$$\therefore \text{P.I.} = \frac{1}{2} \cdot \frac{\cos x}{D^2 + 1} - \frac{1}{2} \cdot \frac{\cos 2x}{D^2 + 1} \text{ (first term case of failure)}$$

$$= \frac{1}{2} x \cdot \frac{\cos x}{6D^3} - \frac{1}{2} \cdot \frac{\cos 2x}{(-4)^2 + 1}$$

$$= \frac{1}{2} x \cdot \frac{\cos x}{6(-1)^2 D} + \frac{1}{126} \cos 2x$$

$$= \frac{1}{12} x \sin x + \frac{1}{126} \cos 2x \text{ as } \frac{1}{D} \text{ means integration}$$

Hence the complete solution is $y = \text{C.F.} + \text{P.I.}$

Ex. 9. Solve $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 2y = e^x + \cos x$.

[Agra 60; Punjab M.A. 57; Vikram 62; Poona 64; Bombay 58]

Solution. Auxillary equation is $D^3 - 3D^2 + 4D - 2 = 0$,
i.e., $(D-1)(D^2 - 2D + 2) = 0$, i.e., $(D-1)[(D-1)^2 + 1] = 0$

or $(D-1)1 \pm i$; $\therefore$ C.F. $= C_1 e^x + e^x (C_2 \cos x + C_3 \sin x)$

$$\text{P.I.} = \frac{1}{D^3 - 3D^2 + 4D - 2} e^x + \frac{1}{D^3 - 3D^2 + 4D - 2} \cos x$$

first term is case of failure

$$= x \frac{1}{3D^2 - 6D + 4} e^x + \frac{1}{(-1)D - 3(-1) + 4D - 2} \cos x$$

$$= x e^x + \frac{1}{3D + 1} \cos x = x e^x + \frac{3D - 1}{9D^2 - 1} \cos x$$

$$= x e^x + \frac{1}{9} (3 \sin x + \cos x).$$

Hence the complete solution is $y = \text{C.F.} + \text{P.I.}$

Ex. 10. Solve $(D^3 - 5D^2 + 7D - 2)y = e^{2x} \cosh x$.

[Delhi Hons. 55]

Solution. A.E. is $D^3 - 5D^2 + 7D - 2 = 0$.

$$(D-1)(D^2 - 4D + 3) = 0 \quad \text{or} \quad (D-1)(D-3)(D-1) = 0.$$

$$\therefore \text{C.F.} = (C_1 + C_2 x) e^x + C_3 e^{3x}.$$

$$\text{P.I.} = \frac{e^{2x} \cosh x}{(D-1)^2(D-3)} = \frac{e^{2x} \cdot \frac{1}{2}(e^x + e^{-x})}{(D-1)^2(D-3)}$$

$$= \frac{1}{2} \frac{e^{3x}}{(D^3 - 5D^2 + 7D - 3)} + \frac{1}{2} \frac{e^x}{D^3 - 5D^2 + 7D - 3}$$

both cases of failure

$$= \frac{1}{2} x \frac{e^{3x}}{3D^2 - 10D + 7} + \frac{1}{2} x \frac{e^x}{3D^2 - 10D + 7}$$

$$= \frac{1}{2} x \frac{e^{3x}}{3 \cdot 3^2 - 10 \cdot 3 + 7} + \frac{1}{2} x^2 \frac{e^x}{6D - 10}$$

$$= \frac{1}{6} x e^{3x} - \frac{1}{6} x^2 e^x.$$

Hence the complete solution is $y = \text{C.F.} + \text{P.I.}$

Ex. 11. Solve $[D^4 + (m^2 + n^2)D^2 + m^2 n^2]y$

$$= \cos \frac{1}{2}(m+n)x \cos \frac{1}{2}(m-n)x.$$

[Delhi Hons. 53]

Solution. A.E. is $(D^2 + m^2)(D^2 + n^2) = 0$, $D = \pm mi, \pm ni$.

$$\text{C.F.} = C_1 \cos(mx + C_2) + C_3 \cos(nx + C_4).$$

$$\text{P.I.} = \frac{\cos \frac{1}{2}(m+n)x \cos \frac{1}{2}(m-n)x}{D^4 + (m^2 + n^2)D^2 + m^2 n^2}$$

$$= \frac{1}{2} \frac{\cos mx + \cos nx}{D^4 + (m^2 + n^2)D^2 + m^2 n^2} \text{ cases of failure}$$

$$= \frac{1}{2} x \frac{\cos mx + \cos nx}{4D^3 + 2(m^2 + n^2)D}$$

$$\begin{aligned}
 &= \frac{1}{2} x \frac{1}{2D} \left[\frac{\cos mx}{-2m^2 + (m^2 + n^2)} + \frac{\cos nx}{+2n^2 + (m^2 - n^2)} \right] \\
 &= \frac{x}{4(m^2 - n^2)} \left[\frac{\cos mx}{D} + \frac{\cos nx}{D} \right] \\
 &= \frac{x}{4(m^2 - n^2)} \left[-\frac{1}{m} \sin mx + \frac{1}{n} \sin nx \right].
 \end{aligned}$$

The complete solution is $y = \text{C.F.} + \text{P.I.}$

✓ 5.18. $\frac{1}{f(D)} x^m$, where m is a positive integer. [Gujrat 59]

$$\begin{aligned}
 \text{Consider } & \frac{1}{D - \alpha} x^m \\
 &= -\frac{1}{\alpha(1 - D/\alpha)} x^m = -\frac{1}{\alpha} \left(1 - \frac{D}{\alpha} \right)^{-1} x^m \\
 &= \frac{1}{\alpha} \left(1 + \frac{D}{\alpha} + \frac{D^2}{\alpha^2} + \dots + \frac{D^m}{\alpha^m} + \dots \right) x^m \\
 &= -\frac{1}{\alpha} \left(\frac{x^m}{\alpha} + \frac{mx^{m-1}}{\alpha^2} + \frac{m(m-1)x^{m-2}}{\alpha^3} + \dots + \frac{m!}{\alpha^{m+1}} \right).
 \end{aligned}$$

Therefore to evaluate $\frac{1}{f(D)} x^m$ expand $[f(D)]^{-1}$ in ascending powers of D , retaining terms as far D^m and operate each term on x^m .

We need not retain terms containing D^{m+1} , D^{m+2} etc. as $D^{m+1}x = 0$, $D^{m+2}x = 0$ etc.

(Ex.) Solve $(D^3 + 2D^2 + D)y = e^{2x} + x^2 + x$. [Poona 64]

Solution. A.E. is $D(D+1)^2 = 0$, i.e., $D = 0, -1, -1$.

$\therefore$ C.F. $= C_1 + (C_2 + C_3 x)e^{-x}$.

$$\begin{aligned}
 \text{P. I.} &= \frac{e^{2x}}{D(D+1)^2} + \frac{1}{D(1+D)^2} (x^2 + x) \\
 &= \frac{e^{2x}}{2(2+1)^2} + \frac{1}{D(1+D)^2} (x^2 + x) \\
 &= \frac{e^{2x}}{18} + \frac{1}{D} [1 - 2D + 3D^2 \dots] (x^2 + x) \\
 &= \frac{e^{2x}}{18} + \frac{1}{D} [x^2 + x - 4x - 2 + 6] \\
 &= \frac{e^{2x}}{18} + \frac{x^2}{3} - \frac{3x^2}{2} + 4x.
 \end{aligned}$$

The complete solution is $y = \text{C.F.} + \text{P.I.}$

• 5.19. To show that $\frac{1}{f(D)} (e^{ax} V) = e^{ax} \frac{1}{f(D+a)} V$, where V is function of x .

[Delhi Honv. 62, 55; Karnatak 61; Bombay 58]

We have on successive differentiation (by parts),

$$D(e^{ax}V) = e^{ax}DV + ae^{ax}V = e^{ax}(D+a)V,$$

$$D^2(e^{ax}V) = e^{ax}D^2V + ae^{ax}DV + a^2e^{ax}V + ae^{ax}DV \\ = e^{ax}(D^2 + 2aD + a^2)V = e^{ax}(D+a)^2V.$$

Similarly, $D^3(e^{ax}V) = e^{ax}(D+a)^3V$
and $D^n(e^{ax}V) = e^{ax}(D+a)^nV.$

Therefore $f(D)(e^{ax}V) = e^{ax}f(D+a)V.$

[Poona 62]

Taking the inverse operators, we have

$$\frac{1}{f(D)}(e^{ax}V) = e^{ax} \frac{1}{f(D+a)}V.$$

Thus we find that operator $\frac{1}{f(D)}$ on $(e^{ax}V)$ is equivalent to $\frac{1}{f(D+a)}$ on V taking e^{ax} outside.

Therefore in practice take out e^{ax} and put $(D+a)$ in place of D and then find $\frac{1}{f(D+a)}V$ as usual.

Ex. 1. Solve $\frac{d^2y}{dx^2} - 9y = 6e^{3x} + xe^{3x}.$

[Bombay 61]

Solution. Auxiliary equation is $D^2 - 9 = 0$, $D = \pm 3$.

$$\text{C.F.} = C_1e^{3x} + C_2e^{-3x}.$$

$$\begin{aligned} \text{P. I.} &= \frac{1}{D^2 - 9} e^{3x} (6+x) = e^{3x} \frac{1}{(D+3)^2 - 9} (6+x) \\ &= e^{3x} \frac{1}{D^2 + 6D} (6+x) = e^{3x} \frac{1}{6D} (1 + \frac{1}{6}D)^{-1} (6+x) \\ &= e^{3x} \frac{1}{6D} (1 - \frac{1}{6}D - \dots) (6+x) \\ &= e^{3x} \frac{1}{6D} (6+x - \frac{1}{6}(6+x)) = \frac{1}{36} e^{3x} (35x + 3x^2). \end{aligned}$$

Hence the complete solution is

$$y = C_1e^{3x} + C_2e^{-3x} + \frac{1}{36}e^{3x} (35x + 3x^2).$$

Ex. 2. Solve $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - y = xe^x + e^x.$

[Agra 61 ; Bombay 58]

Solution. A.E. is $D^3 - 3D^2 + 3D - 1 = 0$,
i.e., $(D-1)^3 = 0$ or $D = 1, 1, 1$.

$$\therefore \text{C.F.} = (C_1 + C_2x + C_3x^2)e^x.$$

$$\text{P. I.} = \frac{1}{(D-1)^3} e^x (x+1)$$

$$\begin{aligned} &= e^x \frac{1}{(D+1-1)^3} (x+1) = e^x \frac{1}{D^3} (x+1) \\ &= e^x \frac{1}{D^3} \frac{(x+1)^2}{2} = e^x \cdot \frac{1}{D} \frac{(x+1)^3}{6} \end{aligned}$$

$$= e^x \cdot \frac{(x+1)^4}{24}$$

Hence the general solution is $y = C.F. + P.I.$

Ex. 3. Solve $(D^3 - 7D - 6)y = e^{2x} \cdot x^2$.

[Bombay B. Sc. 61]

Solution. A.E. is $D^3 - 7D - 6 = 0$;

i.e., $(D+1)(D^2 - D - 6) = 0$ or $(D+1)(D-3)(D+2) = 0$.

$\therefore$ C.F. $= C_1 e^{-x} + C_2 e^{3x} + C_3 e^{-2x}$.

$$\begin{aligned} P.I. &= \frac{e^{2x} \cdot x^2}{D^3 - 7D - 6} = e^{2x} \frac{1}{D+2)^3 - 7(D+2) - 6} x^2 \\ &= e^{2x} \frac{1}{D^3 + 6D^2 + 5D - 12} x^2 \\ &= -\frac{e^{2x}}{12} \left(1 - \frac{5}{12}D - \frac{1}{2}D^2 - \frac{1}{12}D^3 \right)^{-1} x^2 \\ &= -\frac{e^{2x}}{12} \left(1 + \frac{5}{12}D + \frac{1}{2}D^2 + \frac{25}{12^2}D^2 \right) x^2 \\ &= -\frac{e^{2x}}{12} \left(x^2 + \frac{5}{6}x + \frac{97}{72} \right) \text{ etc.} \end{aligned}$$

Ex. 4. Solve $\frac{d^3 y}{dx^3} - 2 \frac{dy}{dx} + 4y = e^x \cos x$.

[Delhi Hons. 54; Karnatak 61]

Solution. Auxiliary equation is $D^3 - 2D + 4 = 0$,

i.e., $(D+2)(D^2 - 2D + 2) = 0$

or $(D+2)[(D-1)^2 + 1] = 0$.

$D = -2, 1 \pm i$, C.F. $= C_1 e^{-2x} + C_2 e^x \cos(x + C_3)$.

$$P.I. = \frac{1}{D^3 - 2D + 4} e^x \cos x$$

$$= e^x \frac{1}{(D+1)^3 - 2(D+1) + 4} \cos x$$

$$= e^x \frac{1}{D^3 + 3D^2 + D + 3} \cos x$$

$$= e^x \cdot x \frac{1}{3D^2 + 6D + 1} \cos x$$

$$= x e^x \frac{1}{-3 + 6D + 1} \cos x$$

$$= x e^x \frac{1}{6D - 2} \cos x$$

$$= \frac{1}{2} x e^x \frac{3D + 1}{9D^2 - 1} \cos x$$

$$= -\frac{1}{2} x e^x (-3 \sin x + \cos x)$$

$$= \frac{1}{2} x e^x (3 \sin x - \cos x).$$

(case of failure)

Hence the complete solution is $y = C.F. + P.I.$

Ex 5. Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 4y = e^x \cos x$.

[Agra 59]

Solution. A.E. is $D^2 - 2D + 4 = 0$

or $[(D-1)^2 + 3] = 0$ or $D = 1 \pm \sqrt{3}i$

$\therefore$ C.F. $= e^x [C_1 \cos \sqrt{3}x + C_2 \sin \sqrt{3}x]$

P.I. $= \frac{1}{D^2 - 2D + 4} e^x \cos x$

$= e^x \frac{1}{(D+1)^2 - 2(D+1) + 4} \cos x$

$= e^x \frac{1}{D^2 + 3} \cos x = e^x \frac{1}{-1 + 3} \cos x$

$= \frac{1}{2} e^x \cos x$, etc.

Ex 6(a) Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = e^x \sin x$.

[Osmania 62]

Solution. A.E. is $D^2 - 2D + 2 = 0$,

i.e., $(D-1)^2 + 1 = 0$ or $D = 1 \pm i$.

C.F. $= e^x [C_1 \cos x + C_2 \sin x]$.

P.I. $= \frac{e^x \sin x}{D^2 - 2D + 2} = e^x \cdot \frac{1}{(D+1)^2 - 2(D+1) + 1} \sin x$

$= e^x \frac{1}{D^2 + 1} \sin x$

(case of failure)

$= e^x \cdot x \frac{1}{2D} \sin x = -\frac{x}{2} e^x \cos x$, etc.

Ex. 6(b) Solve $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = xe^{-x}$.

[Poona 61]

Solution. A.E. is $D^2 + 2D + 2 = 0$, $(D+1)^2 + 1 = 0$,

$D = -1 \pm i$; $\therefore$ C.F. $= C_1 e^{-x} \cos(x + C_2)$

P.I. $= \frac{1}{D^2 + 2D + 2} xe^{-x}$

$= e^{-x} \frac{1}{(D-1)^2 + 2(D-1) + 2} x$

$= e^{-x} \frac{1}{D^2 + 1} x = e^{-x} (1 - D^2 \dots) x$

$= xe^{-x}$, etc. ✓

5 20. $\frac{1}{f(D)} (xV)$, where V is any function of x .

[Poona 60; Karnatak 60, 61, 62]

We have

$D^n(x)V = xD^nV + nD^{n-1}V$ by Leibnitz's rule

$$= x D^n V + \frac{d}{dD} (D^n) V \text{ as } \frac{d}{dD} D^n = n D^{n-1}.$$

$$\therefore f(D) (xV) = x f(D) V + f'(D) V.$$

Taking the inverse operator, we get

$$\frac{1}{f(D)} (xV) = x \frac{1}{f(D)} V + \left[\frac{d}{dD} \frac{1}{f(D)} \right] V.$$

$$\text{or } \frac{1}{f(D)} (xV) = x \frac{1}{f(D)} V - \frac{f'(D)}{[f(D)]^2} V.$$

5.21. $\frac{1}{f(D)} (x^m V)$, where V is some function of x .

Now V can have the following different forms :

1. V has the form x^m , then $x^m V$ becomes x^{m+n} which can be evaluated by the method of § 5.18 P. 76.
2. V has the form e^{ax} , then $x^m V$ becomes $x^m e^{ax}$ which can be evaluated by the method of § 5.19 P. 76.
3. V has the form $\cos ax$ or $\sin ax$, then $x^m V$ becomes $x^m \cos ax$ or $x^m \sin ax$, i.e., it is real or imaginary part of $x^m e^{aix}$, which can be easily evaluated.

Ex. 1. Solve $\frac{d^4 y}{dx^4} - y = x \sin x$.

[Mysore 68; Agra 66; Lucknow Pass 57]

Solution. Auxiliary equation is

$$D^4 - 1 = 0, (D^2 + 1)(D^2 - 1) = 0, D = \pm i, \pm 1.$$

$$\text{Hence C.F.} = C_1 \cos x + C_2 \sin x + C_3 e^x + C_4 e^{-x}.$$

$$\text{P. I.} = \frac{1}{D^4 - 1} x \sin x.$$

$$= \text{Imaginary Part of } \frac{1}{D^4 - 1} x e^{ix}$$

$$= \text{I. P. of } e^{ix} \frac{1}{(D+i)^4 - 1} x$$

$$= \text{I. P. of } e^{ix} \frac{1}{D^4 + 4iD^3 - 6D^2 - 4iD} x$$

$$= \text{I. P. of } -e^{ix} \frac{1}{4iD} [1 - \frac{3}{2}iD - D^2 + \frac{1}{4}iD^3]^{-1} x \left(\because \frac{1}{i} = -i \right)$$

$$= \text{I. P. of } -e^{ix} \frac{1}{4iD} [1 + \frac{3}{2}iD] x$$

$$= \text{I. P. of } -e^{ix} \frac{1}{4iD} [x + \frac{3}{2}i]$$

$$= \text{I. P. of } \frac{i}{4} (\cos x + i \sin x) [\frac{1}{2}x^2 + \frac{3}{2}ix]$$

$$= \frac{1}{8}x^2 \cos x - \frac{3}{8}x \sin x.$$

Hence the complete solution is

$$y = C_1 \cos x + C_2 \sin x + C_3 e^x + C_4 e^{-x} + \frac{1}{8} x^2 \cos x - \frac{1}{8} x \sin x.$$

Ex. (2) Solve $\frac{d^2 y}{dx^2} + 4y = x \sin x$.

[Delhi Hons. 63]

Solution. A. E. is $D^2 + 4 = 0$, $D = \pm 2i$.

$$C. F. = C_1 \cos 2x + C_2 \sin 2x.$$

$$P. I. = \frac{1}{D^2 + 4} x \sin x$$

$$= I. P. \text{ of } \frac{1}{D^2 + 4} x e^{ix}$$

$$= I. P. \text{ of } e^{ix} \frac{1}{(D+i)^2 + 4} x$$

$$= I. P. \text{ of } e^{ix} \frac{1}{D^2 + 2Di + 3} x$$

$$= I. P. \text{ of } \frac{1}{3} e^{ix} (1 + \frac{2}{3} Di + \frac{1}{3} D^2)^{-1} x$$

$$= I. P. \text{ of } \frac{1}{3} e^{ix} (1 - \frac{2}{3} Di) x$$

$$= I. P. \text{ of } \frac{1}{3} (\cos x + i \sin x) (x - \frac{2}{3} i)$$

$$= \frac{1}{9} (3x \sin x - 2 \cos x).$$

The complete solution is $y = C.F. + P.I.$

Ex. 3. (a) Solve $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = 3x^2 e^{2x} \sin 2x$.

[Lucknow Pass 58]

Solution. A. E. is $D^2 - 4D + 4 = 0$, $(D-2)^2 = 0$.

$$\therefore C.F. = (C_1 + C_2 x) e^{2x}.$$

$$P. I. = \frac{3x^2 e^{2x} \sin 2x}{(D-2)^2} = 3e^{2x} \cdot \frac{1}{(D+2-2)^2} x^2 \sin 2x$$

$$= 3e^{2x} \frac{1}{D^2} x^2 \sin 2x$$

$$= I. P. \text{ of } 3e^{2x} \frac{1}{D^2} x^2 e^{2ix}$$

$$= I. P. \text{ of } 3e^{2x} \cdot e^{2ix} \frac{1}{(D+2i)^2} x^2$$

$$= I. P. \text{ of } 3e^{2x} e^{2ix} \frac{1}{D^2 + 4iD - 4} x^2$$

$$= I. P. \text{ of } -3e^{2x} e^{2ix} \cdot \frac{1}{4} (1 - iD - \frac{1}{4} D^2)^{-1} x^2$$

$$= I. P. \text{ of } -3e^{2x} e^{2ix} \cdot \frac{1}{4} (1 + iD + \frac{1}{4} D^2 - D^2) x^2$$

$$= I. P. \text{ of } -3e^{2x} e^{2ix} \cdot \frac{1}{4} (x^2 + 2ix - \frac{3}{2})$$

$$= I. P. \text{ of } -\frac{3}{4} e^{2x} (\cos 2x + i \sin 2x) (x^2 + 2ix - \frac{3}{2})$$

$$= -\frac{3}{8} e^x [(2x^2 - 3) \sin 2x + 4x \cos 2x].$$

The complete solution is $y = C.F. + P.I.$

Proceed as above.

Ex. 3. (b) Solve $(D^2+1)y=8x^2e^{2x}\sin 2x$. [Allahabad 66]

*Ex. 4. Solve $(D^4+2D^2+1)y=x^2\cos x$.

[Delhi Hons. 64, 62; Lucknow Pass 56; Benaras B.Sc. 59;
Ujjain 61, 59; Karnatak 62; Nagpur B.Sc. 55]

Solution. The auxiliary equation is

$$(D^4+2D^2+1)=0 \text{ i.e., } (D^2+1)^2=0,$$

$$D=\pm i, \pm i.$$

$$\therefore \text{C. F.}=(C_1+C_1x)\cos x+(C_3+C_4x)\sin x.$$

$$\text{P. I.}=\frac{1}{D^4+2D^2+1}x^2\cos x=\text{real part of } \frac{1}{(D^2+1)^2}x^2e^{ix}.$$

$$\begin{aligned}\text{But } \frac{1}{(D^2+1)^2}x^2e^{ix} &= e^{ix} \frac{1}{[(D+i)^2+1]^2}x^2 \\ &= e^{ix} \frac{1}{(D^2+2iD)^2}x^2 = e^{ix} \frac{1}{-4D^2(1-\frac{1}{2}iD)^2}x^2 \\ &= -\frac{1}{4}e^{ix} \frac{1}{D^2}(1-\frac{1}{2}iD)^{-2}x^2 \\ &= -\frac{1}{4}e^{ix} \frac{1}{D^2}(1+iD-\frac{3}{4}D^2-\frac{1}{2}iD^3+\frac{5}{16}D^4+\dots)x^2 \\ &= -\frac{1}{4}e^{ix} \frac{1}{D^2}[x^2+2ix-\frac{3}{2}] \\ &= -\frac{1}{4}e^{ix} \left[\frac{x^4}{12} + i\frac{x^3}{3} - \frac{3}{4}x^2 \right] \\ &= -\frac{1}{4}(\cos x + i\sin x) \left[\frac{x^4}{12} - \frac{3}{4}x^2 + i\frac{x^3}{3} \right]\end{aligned}$$

$\therefore$ P.I. = real part of the above

$$= -\frac{1}{4} \left[\left(\frac{x^4}{12} - \frac{3}{4}x^2 \right) \cos x - \frac{1}{3}x^3 \sin x \right]$$

The complete solution is $y = \text{C. F.} + \text{P. I.}$

Ex. 5. (a) Solve $\frac{d^2y}{dx^2} + a^2y = x \cos ax$.

Solution. A. E. is $D^2 + a^2 = 0$, $D = \pm ai$.

$$\text{C.F.} = C_1 \cos ax + C_2 \sin ax.$$

$$\text{P. I.} = \frac{x \cos ax}{D^2 + a^2}$$

$$= \text{Real part of } \frac{xe^{aix}}{D^2 + a^2}$$

$$= \text{R. P. of } e^{ix} \frac{1}{(D+ai)^2 + a^2} x$$

$$= \text{R. P. of } e^{aix} \frac{1}{D^2 + 2aiD} x$$

[Nagpur 57]

$$\begin{aligned}
 &= \text{R. P. of } e^{aix} \frac{1}{2aiD} \left[1 - \frac{1}{ia} iD \right]^{-1} x \\
 &= \text{R. P. of } e^{aix} \frac{1}{2aiD} \left[x + \frac{1}{2a} i \right] \\
 &= \text{R. P. of } -\frac{i}{2a} (\cos ax + i \sin ax) \left[\frac{x^2}{2} + \frac{1}{2a} ix \right] \\
 &= \frac{1}{4a^2} [ax^2 \sin ax + x \cos ax]
 \end{aligned}$$

The complete solution is $y = \text{C. F.} + \text{P. I.}$

Ex. 5. (b) Solve $(D^2 + 4)y = x \cos 2x$. [Karnatak 61]

Hint. Put $a=2$ in the above example.

Ex. 5. (c) Solve $\frac{d^2y}{dx^2} + y = x \cos x$. [Poona 61 (S)]

Hint. Put $a=1$ in the above example.

Ex. 6. (a) Solve $\frac{d^2y}{dx^2} - y = x^2 \cos x$. [Karnatak 60, 61]

Solution. Auxiliary equation is $D^2 - 1 = 0$ i.e. $D = \pm 1$.

$$\therefore \text{C.F.} = C_1 e^x + C_2 e^{-x}.$$

$$\text{P. I.} = \frac{1}{D^2 - 1} x^2 \cos x$$

$$= \text{Real part of } \frac{1}{D^2 - 1} x^2 e^{ix}$$

$$= \text{R.P. of } e^{ix} \frac{1}{(D+i)^2 - 1} x^2$$

$$= \text{R.P. of } e^{ix} \frac{1}{D^2 + 2Di - 2} x^2$$

$$= \text{R.P. of } -e^{ix} \cdot \frac{1}{2} (1 - iD - \frac{1}{2} D^2)^{-1} x^2.$$

$$= \text{R.P. of } -\frac{1}{2} e^{ix} (1 + iD + \frac{1}{2} D^2 - D^2) x^2$$

$$= \text{R.P. of } -\frac{1}{2} (\cos x + i \sin x) (x^2 + 2ix - 1)$$

$$= -\frac{1}{2} [(x^2 - 1) \cos x - 2x \sin x].$$

Hence the complete solution is $y = \text{C.F.} + \text{P.I.}$

Ex. 6. (b) $(D^4 - 1)y = x^2 \sin x$. [Delhi 72]

Proceed as above.

Ex. 7. Solve $\frac{d^4y}{dx^4} + 2 \cdot \frac{d^2y}{dx^2} + y = x^2 \cos^2 x$. [Gujrat B.Sc. 62]

Solution. A.E. is

$$D^4 + 2D^2 + 1 = 0, (D^2 + 1)^2 = 0, m = \pm i, \pm i.$$

$$\therefore \text{C.F.} = (C_1 + C_2 x) \cos x + (C_3 + C_4 x) \sin x.$$

$$\text{Now } x^2 \cos^2 x = \frac{1}{2} x^2 (1 + \cos 2x) = \frac{1}{2} x^2 + \frac{1}{2} x^2 \cos 2x.$$

$$\text{P.I. corresponding to } \frac{1}{2} x^2 = (1 + D^2)^{-2} \cdot \frac{1}{2} x^2$$

$$=(1-2D^2\dots) \frac{1}{2}x^2 = \frac{1}{2}x^2 - 2.$$

P.I. corresponding to $\frac{1}{2}x^2 \cos 2x$

$$= \text{R.P. of } \frac{1}{(1+D^2)^2} \cdot \frac{1}{2}x^2 e^{2ix}$$

$$= \text{R.P. of } \frac{1}{2}e^{2ix} \cdot \frac{1}{[1+(D+2i)^2]^2} x^2$$

$$= \text{R.P. of } \frac{1}{2}e^{2ix} \frac{1}{(D^2+4iD-3)^2} x^2$$

$$= \text{R.P. of } \frac{1}{18}e^{2ix} (1-\frac{4}{3}iD-\frac{1}{3}D^2)^{-3} x^2$$

$$= \text{R.P. of } \frac{1}{18}e^{2ix} (1+\frac{8}{3}iD+\frac{2}{3}D^2-\frac{16}{3}D^3\dots) x^2$$

$$= \text{R.P. of } \frac{1}{18}(\cos 2x + i \sin 2x) [x^2 + \frac{16}{3}ix - \frac{2}{3}x^3]$$

$$= \frac{1}{54} [(3x^2-28) \cos 2x - 16x \sin 2x].$$

The general solution is $y = \text{C.F.} + \text{P.I.}$

Miscellaneous Examples

Ex. 1. Solve $(D^2-5D+6)y = 4e^x + 5$.

[Nagpur 61]

Solution. A.E. is $(D-3)(D-2)=0$.

$$\text{C.F.} = C_1 e^{3x} + C_2 e^{2x}.$$

$$\text{P. I.} = \frac{4e^x + 5e^{0x}}{D^2-5D+6} = \frac{4e^x}{1-5+6} + \frac{5}{6} = 2e^x + \frac{5}{6}.$$

$$\text{Hence } y = C_1 e^{3x} + C_2 e^{2x} + 2e^x + \frac{5}{6}.$$

Ex. 2. Solve $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = e^{2x}$.

[Delhi Hons. 62]

Solution. A.E. is $D^3-6D^2+11D-6=0$

$$\text{or } (D-1)(D^2-5D+6)=0 \quad \text{or } (D-1)(D-2)(D-3)=0.$$

$$\therefore \text{C.F.} = C_1 e^x + C_2 e^{2x} + C_3 e^{3x}.$$

$$\text{P. I.} = \frac{e^{2x}}{D^3-6D^2+11D-6} \text{ case of failure}$$

$$= x \frac{e^{2x}}{3D^2-12D+11} = x \frac{e^{2x}}{12-24+11} = -xe^{2x}.$$

$$\text{Hence } y = C_1 e^x + C_2 e^{2x} + C_3 e^{3x} - xe^{2x}.$$

Ex. 3. Solve $\frac{d^3y}{dx^3} - 3\frac{dy}{dx} + 2y = e^{-2x}$.

Solution. A.E. is $(D-1)(D^2+D-2)=0$

[Poona 61]

$$\text{or } (D-1)(D+2)(D-1)=0.$$

$$\text{C.F.} = (C_1 + C_2 x) e^x + C_3 e^{-2x}.$$

$$\text{P. I.} = \frac{e^{-2x}}{D^3-3D+2} \text{ (case of failure)}$$

$$= x \frac{e^{-2x}}{3D^2-3} = \frac{1}{6} x e^{-2x}.$$

$$\text{Hence } y = (C_1 + C_2 x) e^x + C_3 e^{-2x} + \frac{1}{6} x e^{-2x}.$$

Ex. A. Solve $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = e^{2x} + \sin 2x$. [Luck. Pass 57]

Solution. The A.E. is $D^2 - 4D + 4 = 0$
or $(D-2)^2 = 0$ or $D = 2, 2$.

$\therefore$ C.F. $= (C_1 + C_2x) e^{2x}$.

$$\begin{aligned} \text{P. I.} &= \frac{e^{2x}}{D^2 - 4D + 4} + \frac{\sin 2x}{D^2 - 4D + 4} \text{ first term, case of failure} \\ &= x \frac{e^{2x}}{2D - 4} + \frac{\sin 2x}{-4 - 4D + 4} \text{ first term, again case of failure} \\ &= x^2 \frac{e^{2x}}{2} - \frac{1}{D} \sin 2x \\ &= \frac{1}{2} x^2 e^{2x} + \frac{1}{8} \cos 2x. \end{aligned}$$

Hence $y = (C_1 + C_2x) e^{2x} + \frac{1}{2} x^2 e^{2x} + \frac{1}{8} \cos 2x$.

*Ex. 5. Solve $(D^2 + 1)y = e^{2x} \sin x + e^{x/2} \sin(\frac{1}{2}\sqrt{3}x)$. [Lucknow Pass 60]

Solution. A.E. is $D^2 + 1 = 0$, $(D+1)(D^2 - D + 1) = 0$.

$$D = -1, \frac{1 \pm \sqrt{3}i}{2}. \text{ C.F.} = C_1 e^{-x} + C_2 e^{x/2} \cos(\frac{1}{2}\sqrt{3}x + C_3).$$

P.I. corresponding to $e^{2x} \sin x$

$$\begin{aligned} &= \frac{e^{2x} \sin x}{D^2 + 1} = e^{2x} \cdot \frac{1}{(D+2)^2 + 1} \sin x \\ &= e^{2x} \frac{1}{D^2 + 6D + 12D + 9} \sin x = e^{2x} \cdot \frac{1}{-D - 6 + 12D + 9} \sin x \\ &= e^{2x} \frac{1}{11D + 3} \sin x = e^{2x} \cdot \frac{11D - 3}{-11^2 - 3^2} \sin x \\ &= -\frac{1}{130} e^{2x} (11 \cos x - 3 \sin x). \end{aligned}$$

P.I. corresponding to $e^{x/2} \sin(\frac{1}{2}\sqrt{3}x)$

$$\begin{aligned} &= e^{x/2} \frac{1}{(D + \frac{1}{2})^2 + 1} \sin\left(\frac{\sqrt{3}}{2}x\right) \\ &= e^{x/2} \frac{1}{D^2 + \frac{3}{2}D + \frac{5}{4}} \sin \frac{\sqrt{3}}{2}x \text{ (case of failure)} \\ &= e^{x/2} \cdot \frac{1}{3D^2 + 3D + \frac{3}{4}} \sin \frac{\sqrt{3}}{2}x \text{ differentiating the denominator} \\ &\quad \text{w.r.t. } D \text{ and multiplying by } x \\ &= e^{x/2} \cdot \frac{1}{-3 \times \frac{3}{4} + 3D + \frac{3}{4}} \sin \frac{\sqrt{3}}{2}x \\ &= e^{x/2} \cdot \frac{x}{3} \cdot \frac{1}{D - \frac{1}{2}} \sin \frac{\sqrt{3}}{2}x \\ &= e^{x/2} \cdot \frac{x}{3} \cdot \frac{1}{\frac{1}{2} - \frac{1}{2}} \left(\frac{\sqrt{3}}{2} \cos \frac{\sqrt{3}}{2}x + \frac{1}{2} \sin \frac{\sqrt{3}}{2}x \right) \end{aligned}$$

$$= -\frac{x}{6} e^{x/2} \left(\sqrt{3} \cos \frac{\sqrt{3}}{2} x + \sin \frac{\sqrt{3}}{2} x \right)$$

Thus the P.I. = $-\frac{1}{11} e^{2x} (11 \cos x - 3 \sin x)$

$$-\frac{x}{6} e^{x/2} \left(\sqrt{3} \cos \frac{\sqrt{3}}{2} x + \sin \frac{\sqrt{3}}{2} x \right)$$

The complete solution is $y = \text{C.F.} + \text{P.I.}$

Ex. 6. Solve $(D^4 + D^2 + 1)y = e^{-1/2} \cos \left(x \frac{\sqrt{3}}{2} \right)$

[Luck. Pass 59; Punjab M.A. 56; Raj. M.A. 51]

Solution. The auxiliary equation is $(D^4 + D^2 + 1) = 0$
or $[(D^2 + 1)^2 - D^2] = 0$ or $(D^2 - D + 1)(D^2 + D + 1) = 0$.

When $D^2 - D + 1 = 0$, $D = \frac{1 \pm \sqrt{3}i}{2}$,

and when $D^2 + D + 1 = 0$, $D = \frac{-1 \pm 3i}{2}$.

$\therefore \text{C.F.} = C_1 e^{x/2} \cos \left(\frac{1}{2} \sqrt{3} x + C_2 \right) + C_3 e^{-x/2} \cos \left(\frac{1}{2} \sqrt{3} x + C_4 \right)$

P. I. = $\frac{1}{D^4 + D^2 + 1} e^{-x/2} \cos \left(\frac{1}{2} \sqrt{3} x \right)$

$$= e^{-x/2} \frac{1}{(D - \frac{1}{2})^4 + (D - \frac{1}{2})^2 + 1} \cos \left(\frac{1}{2} \sqrt{3} x \right)$$

$$= e^{-x/2} \frac{1}{D^4 - 2D^3 + \frac{5}{2}D^2 - \frac{3}{2}D + \frac{5}{4}} \cos \left(\frac{1}{2} \sqrt{3} x \right) \text{ case of failure}$$

$$= e^{-x/2} x \cdot \frac{1}{4D^3 - 6D^2 + 5D - \frac{3}{2}} \cos \left(\frac{1}{2} \sqrt{3} x \right)$$

multiplying by x and differentiating the denominator

w.r.t. D

$$= e^{-x/2} x \cdot \frac{1}{4D(-\frac{3}{4}) - 6(-\frac{3}{4}) + 5D - \frac{3}{2}} \cos \left(\frac{1}{2} \sqrt{3} x \right)$$

$$= e^{-x/2} x \cdot \frac{1}{2D + 3} \cos \left(\frac{1}{2} \sqrt{3} x \right) = e^{-x/2} x \cdot \frac{2D - 3}{4D^2 - 9} \cos \left(\frac{1}{2} \sqrt{3} x \right)$$

$$= e^{-x/2} x \cdot \frac{1}{4(-\frac{3}{4}) - 9} \left[-2 \cdot \frac{\sqrt{3}}{2} \sin \left(\frac{1}{2} \sqrt{3} x \right) - 3 \cos \left(\frac{1}{2} \sqrt{3} x \right) \right]$$

$$= -\frac{e^{-x/2} x}{12} [-\sqrt{3} \sin \left(\frac{1}{2} \sqrt{3} x \right) - 3 \cos \left(\frac{1}{2} \sqrt{3} x \right)]$$

$$= \frac{1}{4} e^{-x/2} x \cdot \left[\frac{1}{\sqrt{3}} \sin \left(\frac{1}{2} \sqrt{3} x \right) + \cos \left(\frac{1}{2} \sqrt{3} x \right) \right]$$

The complete solution is $y = \text{C.F.} + \text{P.I.}$

Ex. 7. Solve $(D^4 + 2D^3 - 3D^2)y = x^2 + 3e^{2x} + 4 \sin x$.

Solution. The auxiliary equation is $(D^4 + 2D^3 - 3D^2) = 0$ [Delhi Hons. 62, 66]

or $D^2 (D^2 + 2D - 3) = 0$ or $D^2 (D + 3) (D - 1) = 0$.

$$\therefore D = 0, 0, -3, 1.$$

Hence C.F. = $(C_1 + C_2 x) + C_3 e^x + C_4 e^{-3x}$.

$$\begin{aligned} \text{P. I.} &= \frac{1}{D^2 (D^2 + 2D - 3)} (x^2 + 3e^{2x} + 4 \sin x) \\ &= \frac{3e^{2x}}{20} + \frac{4 \sin x}{-2(D-2)} - \frac{1}{3D^2} (1 - \frac{2}{3}D - \frac{1}{3}D^2)^{-1} x^2 \\ &= \frac{3}{20}e^{2x} - \frac{2(D-2)}{D^2-4} \sin x - \frac{1}{3D^2} (1 + \frac{2}{3}D + \frac{1}{3}D^2 + \frac{4}{9}D^2 + \dots) x^2 \\ &= \frac{3}{20}e^{2x} + \frac{2}{5} (\cos x + 2 \sin x) - \frac{1}{3D^2} (x^2 + \frac{4}{3}x + \frac{14}{9}) \\ &= \frac{3}{20}e^{2x} + \frac{2}{5} (\cos x + 2 \sin x) - (\frac{1}{30}x^2 + \frac{2}{15}x + \frac{14}{27}). \end{aligned}$$

Therefore the complete solution is $y = \text{C. F.} + \text{P. I.}$

Ex. 8. Solve $\frac{d^2 y}{dx^2} - 6 \frac{dy}{dx} + 13y = 8e^{3x} \sin 2x$.

[Agra 1954]

Solution. Auxiliary equation is $D^2 - 6D + 13 = 0$.

$$D = \frac{6 \pm \sqrt{36 - 52}}{2} = 3 \pm 2i.$$

$\therefore$ C. F. = $e^{3x} [C_1 \cos 2x + C_2 \sin 2x]$.

$$\begin{aligned} \text{P. I.} &= \frac{1}{(D^2 - 6D + 13)} 8e^{3x} \sin 2x \\ &= 8e^{3x} \frac{1}{(D+3)^2 - 6(D+3) + 13} \sin 2x \\ &= 8e^{3x} \frac{1}{D^2 + 4} \sin 2x \text{ (case of failure)} \\ &= 8e^{3x} \cdot \frac{1}{2D} \sin 2x \\ &= -2xe^{3x} \cos 2x. \end{aligned}$$

Hence the complete solution is

$$y = e^{2x} (C_1 \cos 2x + C_2 \sin 2x) - 2e^{3x} \cos 2x.$$

Ex. 9. (a) If $(D+b)^4 y = \cos ax$, show that the complete solution is $y = (C_1 + C_2 x + C_3 x^2 + C_4 x^3) e^{-bx} + \frac{\cos [bx - 4 \tan^{-1} (a/b)]}{(a^2 + b^2)^2}$.

[Bombay 1961]

(b) Solve $(D^4 + 7D^3 - 3D^2) y = x^2 + 3e^x + 4 \sin x$. [Delhi 1972]

Ex. 10. Solve $\frac{d^3 y}{dx^3} - 2 \frac{d^2 y}{dx^2} - 19 \frac{dy}{dx} + 20y = xe^x + 2e^{-1x} \sin x$.

[Delhi Hons. 1956]

Solution. Auxiliary equation is $(D^3 - 2D^2 - 19D + 20 = 0$

or $(D-1)(D^2 - D - 20) = 0$ or $(D-1)(D-5)(D+4) = 0$,

$\therefore D = 1, 5, -4$, C. F. = $C_1 e^x + C_2 e^{5x} + C_3 e^{-4x}$.